

Duy Pham

 (347)-940-0140  Email  LinkedIn  Github  Portfolio

Education

University of Rochester

Rochester, NY

Master & Bachelor of Science in Computer Science – GPA: 3.6/4.0, Dean's List

May 2027

Relevant coursework: Artificial Intelligence, Design & Analysis of Efficient Algorithms, Software Design, Computer Organization, Database Systems, Computer Networks

Experience

Viettel Networks

June 2025 – August 2025

Software Engineering Intern

Hanoi, VN

- Built and deployed 8 RESTful APIs (Java, Spring Boot, MySQL) for user and bot permission management on the NetMind platform with role-based access control and input validation, serving 5,000+ daily users across 4 bots
- Integrated a centralized logging stack (Fluentd, Elasticsearch, Kibana, Docker) across backend services, parsing log levels into searchable fields for Kibana dashboards and alerts, cutting debugging time by ~40% for 40+ engineers
- Optimized bot and user data queries over 10K+ records, improving backend performance and reducing response times by ~30%, enabling faster user lookups and smoother bot interactions across the platform

VNPT IT

June 2024 – August 2024

Software Engineering Intern

Quang Binh, VN

- Collaborated with a 5-person team to improve a project management system (C++, MySQL, SQLite) tracking 500+ employee records and 50+ active project plans, used daily by 6 departments to coordinate schedules and resources
- Optimized 10+ SQL queries and report generation workflows in C++, enabling faster and more accurate project tracking and reducing report generation time by ~25% for project managers reviewing weekly status updates
- Performed end-to-end database maintenance across 20+ tables, including data migration, automated daily backups, and schema updates, ensuring system reliability, data security, and consistent availability for all teams

The Learning Center

November 2023 – Present

CS & Math Tutor

Rochester, NY

- Tutored 100+ students in CS and Math courses, covering topics such as Java, object-oriented programming, and dynamic programming, and helping students master class material and raise exam scores by ~30%

Projects

Voyagent: Multi-Agent AI Travel Planner

[GitHub](#)

- Built a travel planner where an orchestrator coordinates 5 parallel specialist agents (flights, hotels, weather, activities, pricing) via asyncio, turning budget, dates, and interests into day-by-day itineraries with booking links to 10 sites
- Integrated real-world data through tool calling and web search (Open-Meteo, Wikipedia, OpenStreetMap, ECB, SerpApi), with a budget feedback loop that re-runs flight/hotel searches under price caps when plans exceed budget
- Improved output reliability with self-correcting Pydantic schema validation, code-computed budget totals, and filtering of unverified source links, backed by offline tests that run the full pipeline on scripted model responses

DAYS Distributed Version Control System (Github)

- Designed and implemented a modular distributed version control system in a 4-person team, applying information hiding and abstraction principles to manage over 2,000 lines of code across 8 core modules
- Implemented and tested full functionality for 14 DVCS commands (`init`, `add`, `commit`, `merge`, `push`, `pull`, etc.), achieving 100% correctness across 25+ integration tests and ensuring robust repository synchronization

Technical Skills

Languages: Java, Python, JavaScript, C/C++, Kotlin, Rust, SQL, R, HTML/CSS

Frameworks & Libraries: Spring Boot, Django, Flask, React.js, Pytorch, Scikit-learn, Pandas, FastAPI, RestAPI

Cloud & Databases: Azure, AWS, MySQL, Elasticsearch, MongoDB

Tools: Git, Docker, Kubernetes, Fluentd, Kibana, Linux, Postman, Maven